

Solved Papers	Maharashtra HSC Exam. Topper's Answerwers March 2015	Biology
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Time: 3 Hours

Maximum Marks: 70

Note:

- 1) All questions are compulsory.
- 2) Answer to section-I and Section-II should be written in TWO separate Answerbook.
- 3) Question from Section-I attempted in the Answerbook of section-II and vice-versa will not be assessed and not given any credit.
- 4) Draw neat and well labelled diagram wherever necessary.
- 5) Figures to the right indicate full marks.
- 6) Answer to every new question must begin on a new page.

SECTION – 1 (BOTANY)

1. Select and write the most appropriate Answer form the given alternatives for each sub-question: [7]

(i) A cross used to verify the unknown genotype of F_1 hybrid is a.....

- (a) test
- (b) back
- (c) dihybrid
- (d) monohybrid

Answer. Test

(ii) Appearance of new combinations in F_2 generation in a dihybrid cross proves the law of.....

- (a) dominance
- (b) Segregation
- (c) Independent assortment
- (d) Purity of gametes

Answer. Independent assortment

(iii) A fine powder of recycled modified plastic is known as.....

- (a) polyblend
- (b) polythene
- (c) polyester
- (d) polymer

Answer. Polyblend

(iv) The partially decomposed organic matter is formed by a process called_____.

- (a) Fragmentation
- (b) Humification
- (c) Mineralisation
- (d) Leaching

Answer. Humification

(v) Transfer of genetic material into a bacterial cell through virus vector is known as.....

- (a) Transformation
- (b) Transduction
- (c) Transfection
- (d) Translation

Answer. Transduction

(vi) If the number of chromosomes in an endosperm cell is 27, what will be the chromosome number in the definitive nucleus?

- (a) 9
- (b) 18
- (c) 27
- (d) 36

Answer. 18

(vii) Lever mechanism of pollination is observed in _____.

- (a) Salvia
- (b) Jasmine
- (c) Bougainvillea
- (d) Butea

Answer. Salvia

Q.2 . (A) Answer in ONE sentence only. (6)

- (i) A pea plant pure for yellow seed colour is crossed with a pea plant pure for green seed colour. In F₁ generation, all pea plants were with yellow seeds. Which law of Mendel is applicable?

Answer. “Law of Dominance” or Mendel’s first law of inheritance is applicable.

- (ii) Name the enzyme responsible for delay in ripening of tomato fruits.

Answer. Enzyme “Polygalactourance” is responsible for delay in ripening of tomato fruit.

- (iii) Which are the bacteria responsible for converting organic acids into methane?

Answer. “Methanogenic bacteria” – namely methanococci and methanobacilli are responsible for converting organic acids into methane.

- (iv) Name the high yielding semi-dwarf varieties of wheat selected and introduced in India in 1963.

Answer. “Sonalika” and “Kalyansona” are semidwarf varieties of wheat selected and introduced in India in 1963.

- (v) “Formation of primary endosperm nucleus is called triple fusion’. Give reason.

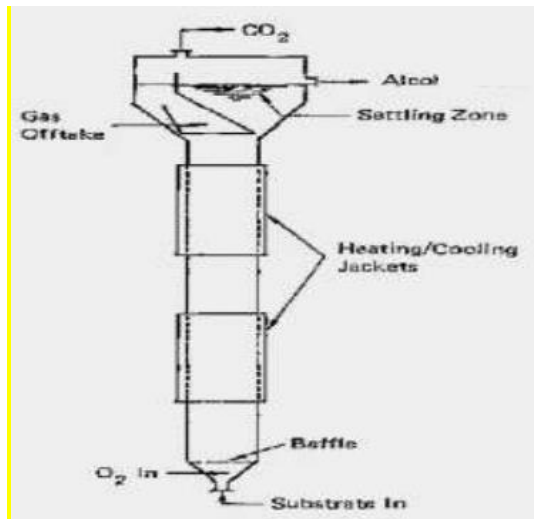
Answer. Formation of PEN (primary endosperm nucleus) involves fusion of secondary nucleus, which is union product of two polar nuclei and one male gamete.

- (vi) Mention any 'two' ecological services for benefit of mankind.

Answer. CO₂ fixation, release of oxygen and pollination are the ecological services for benefit of mankind.

Q .2 (B) Sketch and label: 'Tubular tower fermenter'.

Answer. TUBULAR TOWER FERMENTER



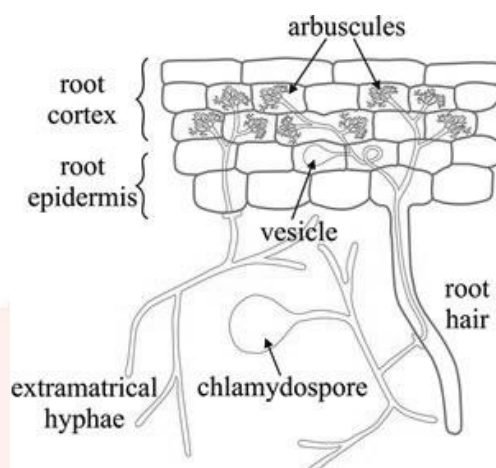
Q2 (C) Attempt any two of the following:

- (i) Global warming is caused by greenhouse effect. Justify

Answer: The greenhouse is a small glass house used for growing plants during winter. The glass panel lets the light in but does not allow the heat to escape. Therefore, the greenhouse warms up. Greenhouse effect is a natural phenomenon responsible for heating of Earth's surface and atmosphere which results in global warming.

- (ii) With the help of a neat and labelled diagram explain VAM (Vesicular arbuscular mycorrhizae).

Answer: VAM (Vesicular arbuscular mycorrhizae)



- a. One of our first microbiological products is Vesicular arbuscular mycorrhizae.
 - b. VAM is a fungus that penetrates the roots of a vascular plant in order to help them to capture nutrients from the soil.
 - c. VAM are endomycorrhizae (i.e fungal hyphae growing inside the cortex cells of roots of pteridophytes, orchids, gymnosperm).
 - d. Fungal hyphae penetrate cells and form fine branches or arbuscules, it form enlarged vesicles, hence called VAM.
- (iii) Distinguish between light and dark reactions.

Answer: The main differences between light and dark reaction are:

- 1) Light reaction only takes place in presence of light whereas dark reaction takes place with the presence or absence of sunlight.
- 2) Light reaction is considered to be the light dependent reaction whereas dark reaction is not dependent on light.
- 3) Light reaction is also called photochemical phase whereas dark reaction is also known as biosynthetic phase.
- 4) The end products of light reaction are ATP and NADPH which are also known as assimilatory powers however the end product of dark reaction is called glucose.

5) In light reaction the water molecules split and oxygen is released where as in dark reaction the carbon dioxide is utilised and glucose is produced.

(iv) ‘ Mendel selected the garden pea plant for his experiments’- explain

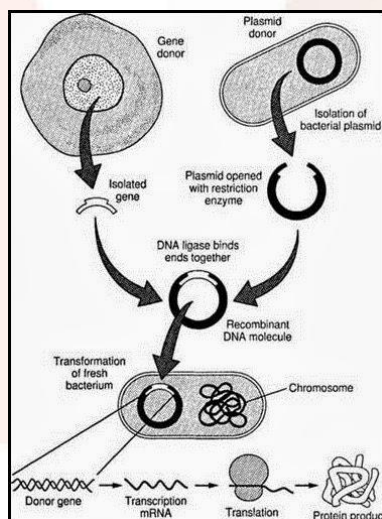
Answer: Gregor Mendel choose pea plants for his experiments because:-

- They are easy to raise, have many offspring per mating, can fertilize, themselves and have varieties in genotype and phenotype that are easily observable.
- Small herbaceous plant with many seeds.
- Self-pollinating plant, which can be cross fertilized artificially.
- Its flowers are large foe easy emasculation.

Q 3 (A) Answer any TWO of the following: [6]

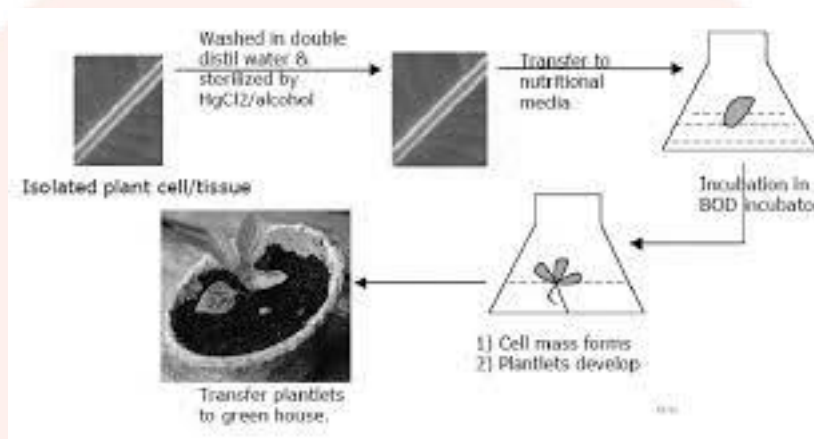
(i) With the help of neat and labelled diagram, describe steps in recombinant DNA technology

Answer:



(ii) What is 'tissue culture'? Describe the methodology of tissue culture.

Answer: Tissue culture – Tissue culture is the growth of tissues or cells separate from the organism. This is typically facilitated via use of a liquid, semi solid or solid growth medium such as broth or agar. Tissue culture commonly refers to the culture of animal cells and tissues with the more specific fern plant tissue culture being used for plants



Methodology of tissue culture:-

I) Explant Culture:

- Explant is either a cell or tissue or piece of plant organ excised
- Explant may be meristem or parenchyma
- Explant is sterilized and place on a solid nutrient medium, where it absorbs nutrients and multiplies.

2) callus formation and its culture:-

- Explant forms unorganized mass of cells or callus. It has thin walled parenchyma cells.
- Cells divide by mitosis, cells are identified

3) Organogenesis:-

- Auxins, cytokinins (growth hormones) are provided to callus in proper proportion
- If auxins are more, roots are formed. If cytokinins are more shoots are formed

(iii) Define vegetative propagation. Differentiate between 'stem tuber' and 'tuberous root'

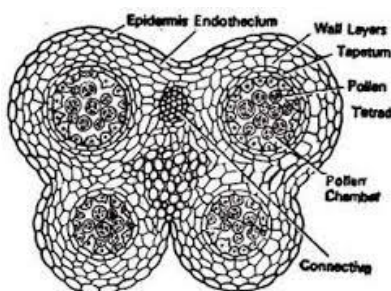
Answer: Vegetative propagation is a type of asexual reproduction of a plant where only a single plant is involved. Parts (stems, roots, leaves and buds) of plant by the process (cutting, grafting and layering) can form a new plant which is genetically identical to its parent plant.

Difference between stem tuber and Tuberous root.

S.No.	Stem Tuber	Tuberous root
1	It is swollen underground stem	It is a swollen root
2	Apical bud is present	Apical bud is absent
3	Presence of nodes and internodes	Nodes and internodes are absent
4	On exposure to sunlight stem tubers may develop chlorophyll	No development of chlorophyll
5	Scaly leaves and axillary buds are present	Scaly leaves and axillary buds are absent

Q 3 (B) Sketch and label: T.S. of angiospermic anther (3)

Answer. T.S. of angiospermic anther:

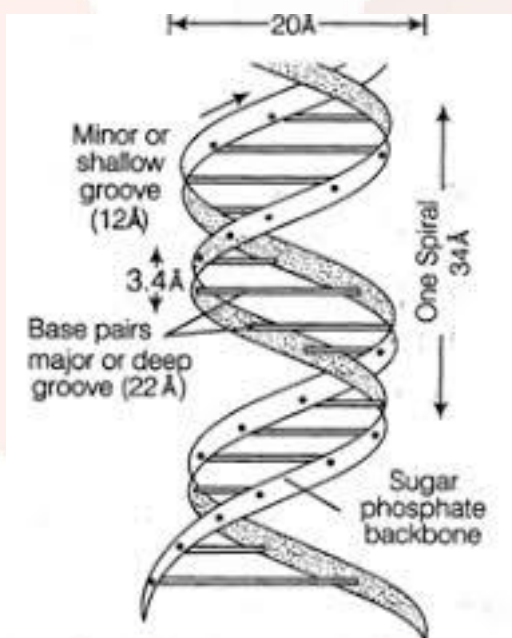


Q.4. With the help of neat and labelled diagram describe Watson and Crick's model of DNA.

Answer. Watson and Crick model of D.N.A:-

- i) Watson and Crick (1963) proposed a structure of DNA. According to them a DNA molecule consists of two anti parallel strand of polynucleotide chains called double helix.
- ii) Structure of each strand:- It has back bone consisting of alternating sugar and phosphate group. Nitrogen bases do not participate in strand formation, they remain attached laterally.
- iii) Base pairing:- Purines : Pyrimidines
 Adenine = Thymine (A=T)
 Guanine= Cytosine(G=C)

Base sequence present on one strand of DNA decides the base sequence of other strand, hence are complementary strands. Between Adenine and Thymine two weak hydrogen bonds are present, while between Guanine and Cytosine three weak hydrogen bonds are present.



(iv) Purine- Pyrimidine ratio: - Purines and pyrimidines ratio is 1:1. This is called Chargaff's rule. It may be present as follows:

$$A+G = T+C = \frac{A+G}{T+C} = 1$$

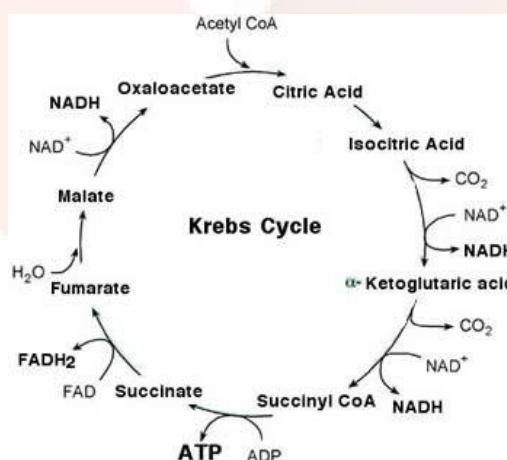
v) Polarity of strand:- One strand of DNA is oriented 3° to 5° and complementary strand is oriented 5° to 3° manner.

vi) Dimensions:- Diameter of DNA molecule is 2nm or 20A° complete spiral, 34A° Distance between two base pairs is 34A°

vii) Major and minor grooves- major groove is 22A° and minor groove = 12A°

Q.4. With the help of schematic representation explain Kreb's cycle.

Answer: The Krebs cycle (or citric acid cycle) is a part of cellular respiration. It is named after HANS KREBS, and is a series of chemical reactions used by all aerobic organisms to generate energy. Its importance to many biochemical pathways suggests that it was one of the earliest parts of cellular metabolism to evolve. The Krebs cycle comes after the link reaction and provides the hydrogen and electrons needed for the electron transport chain. It takes place inside mitochondria.



SECTION –II (ZOOLOGY)

Q5. Select and write most appropriate Answer from the given alternative for each sub-question: [7]

(i) In gene therapy, DNAase is used to treat.....

- | | |
|-------------------------------|---------------------------------------|
| (a) Cystic fibrosis | (b) Haemophilia |
| (c) pituitary dwarfism | (d) insulin dependent diabetes |

Answer. (a) Cystic fibrosis

(ii) If only one X-chromosome is found in female person, which of the following symptoms she will show?

- | | |
|--|--|
| (a) Epicanthal skin fold | (b) Webbing of neck |
| (c) Small testis and absence of spermatogenesis | (d) Presence of simian crease on the palm |

Answer. (b) Webbing of neck

(iii) In which of the following is haploid cells a whole genome in human being is present?

- | | |
|------------------------|---------------------------------|
| (a) Sperm | (b) Somatic cell |
| (c) Mature RBCs | (d) Primary spermatocyte |

Answer. (a) Sperm

(iv) Which of the following is not an example of connecting link?

- | | |
|--------------------------|-----------------------------|
| (a) Archaeopteryx | (b) Ichthyostegia |
| (c) Seymouria | (d) Biston betularia |

Answer. (d) Biston betularia

(v) In....., superior males of one breed are mated with superior females of another breed.

(a) Out crossing

(b) cross breeding

(c) Out breeding

(d) in breeding

Answer. (b) Cross breeding

(vi) Normal activities of the heart are regulated by.....

(a) Brain

(b) spinal cord

(c) modified cardiac muscles

(d) hormones

Answer. (c) Modified cardiac muscles

(vii) During.....type of interaction both organisms are benefited.

(a) Mutualism

(b) competition

(c) Commensalism

(d) parasitism

Answer. (a) mutualism

Q.6. (A) Answer in ONE sentence only: [6]

(i) What is gene pool?

Answer. “The sum total of genes of all individuals of interbreeding population is called gene pool”.

(ii) Give any 2 names of X-linked diseases.

Answer. X- linked disease are – (1) Colour blindness (2) Haemophilia

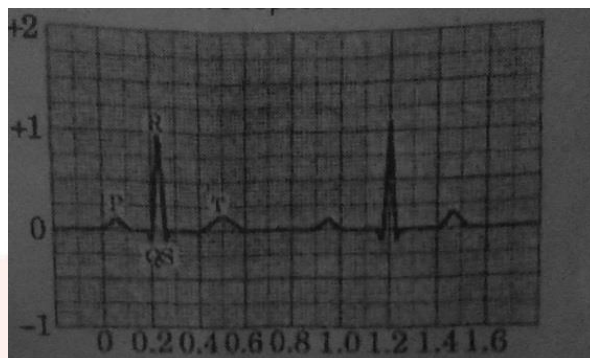
(iii) Abbreviation HGP stands for “Human Genome Project”

Answer: Yes, HGP stand for Human Genome Project.

(iv) Give two varieties of silk which are considered as inferior quality.

Answer. Inferior quality silk are – Tussar silk and Eri silk

(v) In the electrocardiogram shown beside, which wave represents ventricular diastole?



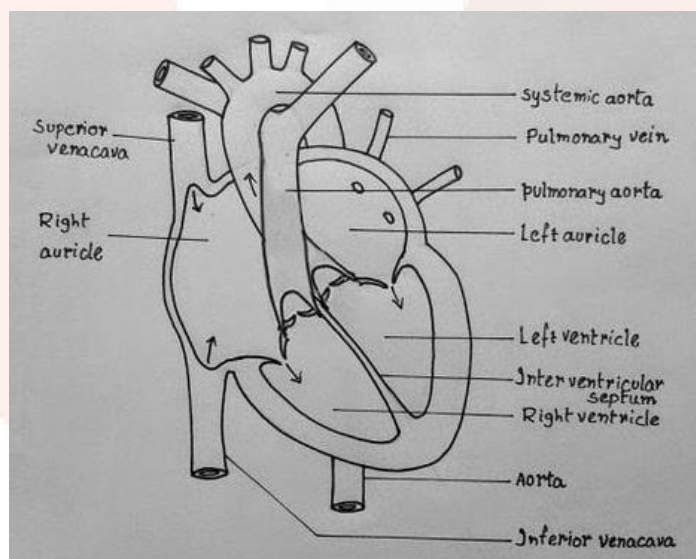
Answer. Wave represent ventricular diastole in the electrocardiogram shown beside.

(vi) Which method of conservation of biodiversity includes hot spot method?

Answer. 'In situ conservation method' include hot spot method.

Q 6 (B) Sketch and label ventral view of human heart. [2]

Answer:



Q 6 (C) Answer any TWO of the following: [4]

(i) Distinguish between Ape and Man

Answer: The difference between Ape and Man are:

- a) Human are said to be the evolutionary products of Apes.
- b) Apes were omnivorous mammals who lived in large jungle areas within trees where as human are mammals who live in cities and towns across the world.
- c) Physically apes were covered in hair and their brain was not as developed as that of human; although the Apes and human share similar organ structures and muscular systems.
- d) Apes used to communicate through grunting and growls where as human communicate through languages.

(ii) Give the application of DNA fingerprinting technique.

Answer: Applications:- DNA fingerprinting technique-

Paternity and Maternity- Because a person inherits his or her VNTs from his or her parents, VNTR patterns can be used to establish paternity and Maternity.

- Criminal investigation and Forensics
- Used to settle insurance claims
- Used to personal identifications
- Used in study of agricultural sciences
- Used in genomics.

(iii) Enlist various types of cancer.

Answer: There are five broad groups that are used to classify cancer.

- 1) Carcinoma: These are characterised by cells that cover internal. In Bone, cartilage, fat, connective tissue, muscles and other supportive tissues.
- 2) Lymphomas: These are cancer that being in the lymph nodes and tissues of immune system.
- 3) Leukemia: These are cancer that begin in the bone marrow and often accumulate in the blood stream.
- 4) Adenoma: These are cancer that arises in the thyroid, the pituitary gland, the adrenal gland and the other glandular tissues.

(iv) Give significance of fertilization

Answer: The significance of fertilization are:

- 1) Fertilization ensures diploid of the organism by fusion of haploid male and female gametes.
- 2) Fertilization provides new genetic constitution to the zygote.
- 3) Fertilization process increases the metabolic activities and the rate of protein synthesis of the egg.

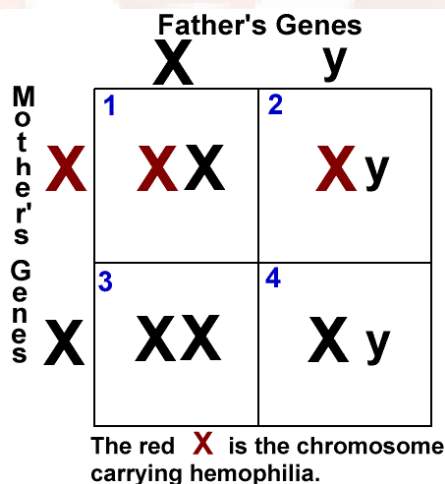
Q7 (A) Answer any TWO of the following:

[6]

(i) What will be phenotype of progeny, if a carrier haemophilic female marries a normal male? Explain.

Answer: If a carrier haemophilic female ($X^H X^h$) marries a male with normal clotting of blood ($X^H Y$) then—

- a) All daughters will have normal clotting of blood, but half of them will be carrier for the disease.
- b) Half the sons will be haemophilic while remaining will have normal clotting of blood.



(ii) With the help of a chart explain the compatibility of human blood groups.

Answer: The two main blood group systems are the ABO system and the Rh(D) type.

- ABO system: In the ABO system, people can be of any of the four groups - O, A, B or AB.

- Rh(D) system: In the Rh (D) system, there can be either Rh(D) positive or Rh(D) negative.

Each system is inherited independently of the other. So, there are eight main blood groups. The chart given below shows the compatibility of different blood groups.

		Donor							
Recipient		O-	O+	B-	B+	A-	A+	AB-	AB+
AB+									
AB-									
A+									
A-									
B+									
B-									
O+									
O-									

(iii) Justify the following sentences:

(a) The conservation of endangered species of plant and animal is necessary.

Answer: The need for conservation of endangered species:

- Laws of habitat
- Disturbance of natural balance
- Maintain ecological process and life support system
- Essential for economic and racial development

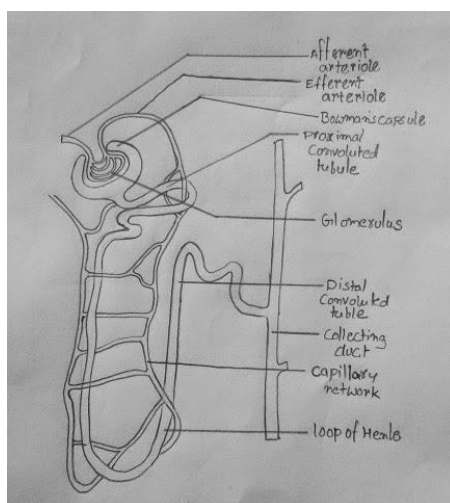
(b) Pollution Under Control (PUC) certificate is mandatory for all vehicles and industries

Answer: Pollution Under Control (PUC) certificate is mandatory for all vehicles and industries because:

- i) Heat generated due to vehicle and Industries leads to global warming,
- ii) Due to incomplete combustion of fuel, hazardous gases like CO hydrocarbons, nitrous oxide are produced which increases air pollution
- iii) Industrial pollution increases SO₂ in air. It damages materials and property causes discolouration and physical deterioration of building materials and sculptures.
- iv) It allows affects vegetation and is capable of irritating respiratory system of animals and human.

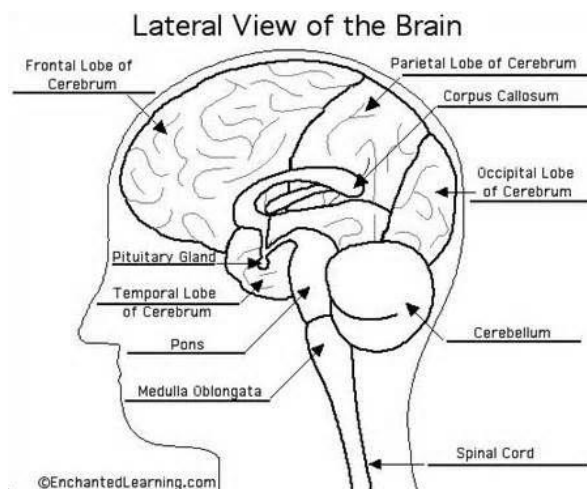
Q7 (B) Sketch and label "Structure of Nephron."

Answer:



Q.8 With the help of labelled diagram of lateral view of cerebrum, describe the structure and give any two functions of cerebrum.

Answer:



(i) The cerebrum is a large part of the brain containing the cerebral centre, as well several sub cortical structures, as well as several sub cortical structures, including the hippocampus, basal ganglia and olfactory bulbs.

ii) In human,, the cerebrum is the uppermost region of the central nervous system.

iii) The cerebrum is divided into approximately symmetric left and right cerebral hemisphere

iv) Two cerebaral hemisphere are interconnected by thick band of transverse nerve fibres called corpus collasum.

v) Such hemisphere has a cavity filled with C & F, called as lateral ventricle or 1st and 2nd ventricle.

(vi) Each cerebral hemisphere is made up of outer cortex and inner medula, Cerebral cortex is made up of grey matter and cerebral medulla is made up of white matter.

(vii) Suter present in cerebrum-

- a) Central suture- Separates frontal lobe from partial lobe
- b) Parietal- Occepital sutuer - Separate parietal lobe from occipital lobe
- c) Lateral or sylvian suture- Separates parietal lobe from temporal lobe.

(viii) Basal ganglia are deep extension of grey matter into white matter of cerebral hemisphere.

Functions of Cerebrum:-

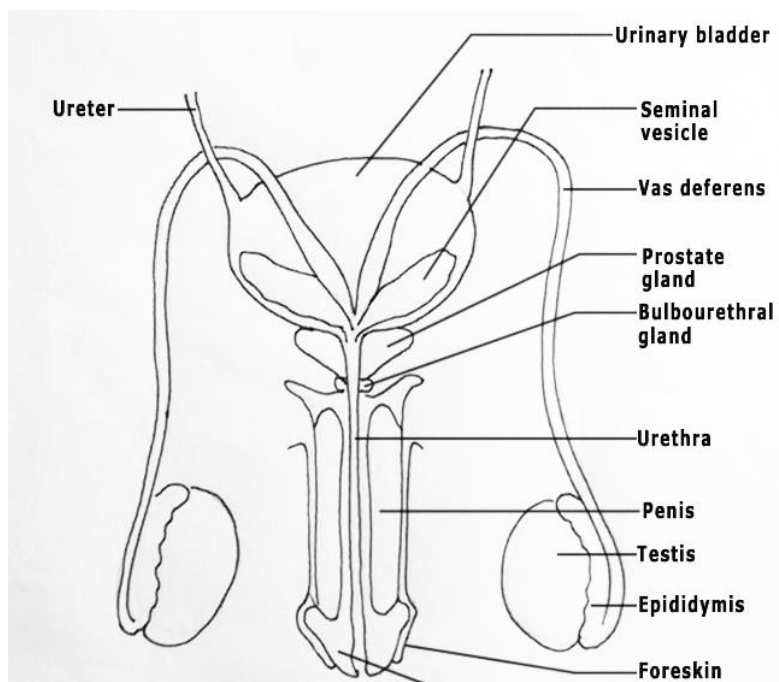
- 1) Cerebrum is a major part of the brain, controlling emotions, hearing,, visions, personality and much more. It controls all voluntary actions.

- 2) Located in frontal lobe- controls association between visions sensations and movements.
It is related with memory.

OR

Q.8 With the help of a neat, labelled diagram, describe the human male reproductive system

Answer:-



Human Male Reproductive System